Intro to Linear Transformations

Idea: Study linear operators that transform spaces of functions into new function spaces.

Example: consider $A: \mathbb{R}^n \to \mathbb{R}^m$, where

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

$$A \qquad x \qquad y$$

$$A(x + \alpha z) = Ax + \alpha Az$$

Given vector spaces V, W, a map $T: V \to W$ is linear (a linear transformation) if

$$T(\alpha f + \beta g) = \alpha Tf + \beta Tg$$

$\in V \quad \in V \qquad \in W \qquad \in W$

$$T(0) = T(0 \cdot f) = 0 \cdot T(f) = 0$$

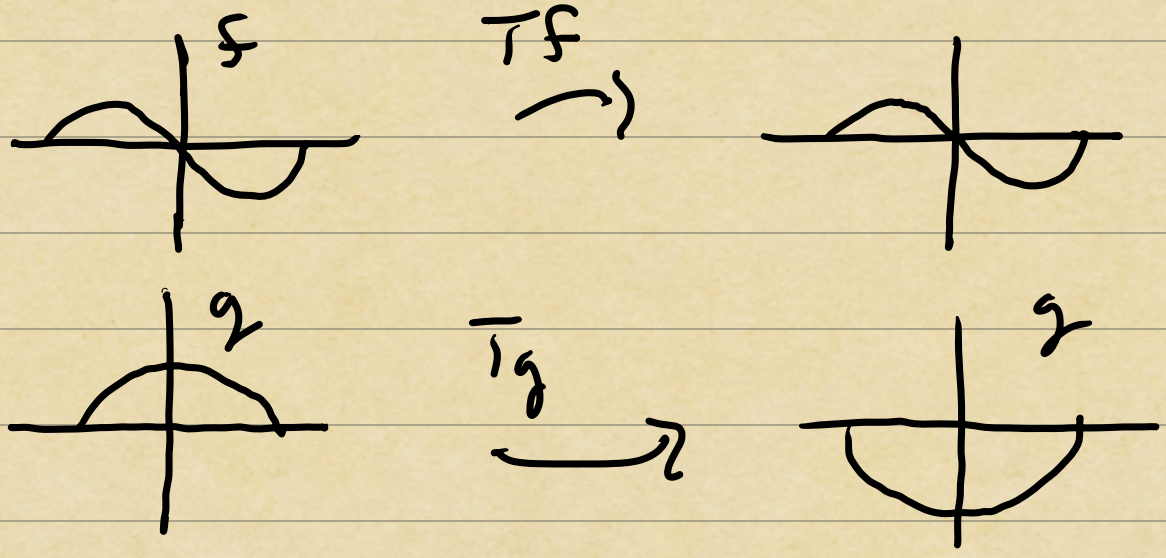
Example: Differentiation on $C^1[-1,1]$

$$\frac{d}{dx}\left(\alpha f(x) + \beta g(x)\right) = \alpha \frac{d}{dx} f(x) + \beta \frac{d}{dx} g(x)$$

Q: What is W (Codomain/Range)? (?)

$\Rightarrow$ Codomain is $C[-1,1]$

Example: $[Tf](x) = -f(-x)$ $\quad C[-1,1]$



Example: Integration (definite) on $C[-1,1]$ (real-valued)

$$I(f) = \int_{-1}^{+1} f(x)\,dx$$

$$I : C([-1,1]) \to \mathbb{R}$$

Example: Integration (indefinite) on $C[-1,1]$

$$[Tf](y) = \int_{-1}^{y} f(x)\,dx \qquad T : C[-1,1] \to C'[-1,1]$$

$$[Df](x) = f'(x) \qquad D : C^1[-1,1] \to C[-1,1]$$

$$DT = I \qquad (D = T^{-1} \text{ "left inverse"})$$

$$TD \overset{?}{=}$$

$$D(x^2+2) = 2x \to T(2x) = x^2\Big|_{-1}^{y} = y^2 - 1$$

$$\downarrow$$

$$Df = g$$

Example: Integral operators on $C[-1,1]$.

$$[Tf](x) = \int_{-1}^{+1} K(x,y) f(y)\, dy$$

$$T(f+\alpha g) = \int_{-1}^{+1} K(x,y)(f(y)+\alpha g(y))\, dy$$

$$= \int_{-1}^{+1} [K(x,y) f(y) + \alpha K(x,y) g(y)]\, dy$$

$$= \int_{-1}^{+1} K(x,y) f(y)\, dy + \alpha \int_{-1}^{+1} K(x,y) g(y)\, dy$$

$$= T(f) + \alpha T(g)$$

onto $\text{span}\{e_1, \dots, e_N\}$

Example: Orthogonal Projection onto a (real) Hilbert Space w/ inner product $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{R}$.

$$f \in H \to P_N f = \sum_{j=1}^{N} \langle f, e_j \rangle e_j$$

Dictionary Approximation

Suppose we want to approximate $f \in C(\Omega)$, $\Omega \subset \mathbb{R}^d$, but we only have samples at data points:

$$f_1 = f(x_1), \dots, f_m = f(x_m)$$

We can choose a dictionary $E = \begin{bmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{bmatrix}$

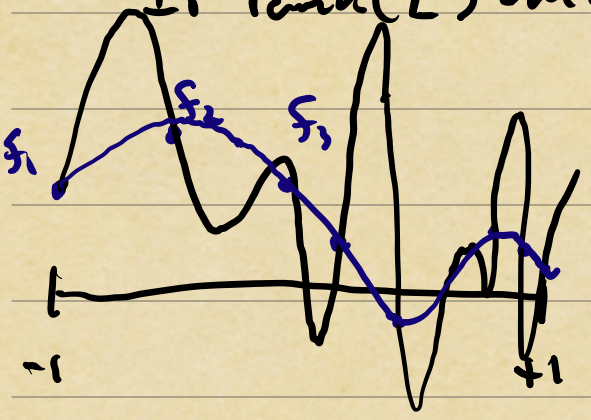
$$f(x) \approx \overset{?}{c_1} e_1(x) + \cdots \overset{?}{c_n} e_n(x)$$

$$\underbrace{\begin{bmatrix} e_1(x_1) & \cdots & e_n(x_1) \\ \vdots & & \vdots \\ e_1(x_m) & \cdots & e_n(x_m) \end{bmatrix}}_{\tilde{E}_{m \times n}} \underbrace{\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}}_{\underline{c}_{n \times 1}} \approx \underbrace{\begin{bmatrix} f_1 \\ \vdots \\ f_m \end{bmatrix}}_{\underline{f}_{m \times 1}}$$

If $n = m = \text{rank}(\tilde{E})$, $\hat{f}$ interpolates data

If $\text{rank}(\tilde{E}) = n < m$, find best fit to data (regression)

If $\text{rank}(\tilde{E}) = m < n$, find suitable solution from potentially many candidates



Polynomial Interpolation ($n = m$, $d = 1$)

Given distinct points $x_0, \dots, x_n \in [-1,1]$, every continuous function $f : [-1,1] \to \mathbb{R}$ has a unique polynomial interpolant of deg $\leq n$:

$$p = P_x f$$

pf] Induction argument for $\det(\tilde{E}_n)$

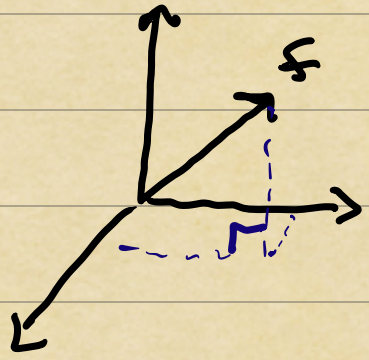
$\Rightarrow$ The map P_x is linear (good ex.)

$\Rightarrow$ The map P_x is a projection ($P_x(P_x f) = P_x f$)

"Best"

$E_{\underline{c}} = P_E \underline{f}$

$\hat{\underline{c}}$ orthogonal projection



"Independent"

$E_{\underline{c}} = P_x \underline{f}$

$\hat{\underline{c}}$ oblique projection

